

Improved Personalized Headline Generation via Denoising Fake Interests from Implicit Feedback

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😊 Background



A thousand readers, a thousand Hamlets.

- ◆ Personalized news headline generation involves generating a *user-specific headline* based on the *user's reading interest*.

- ◆ The key to personalization lies in accurately capturing the *user's true interests*.



News Article

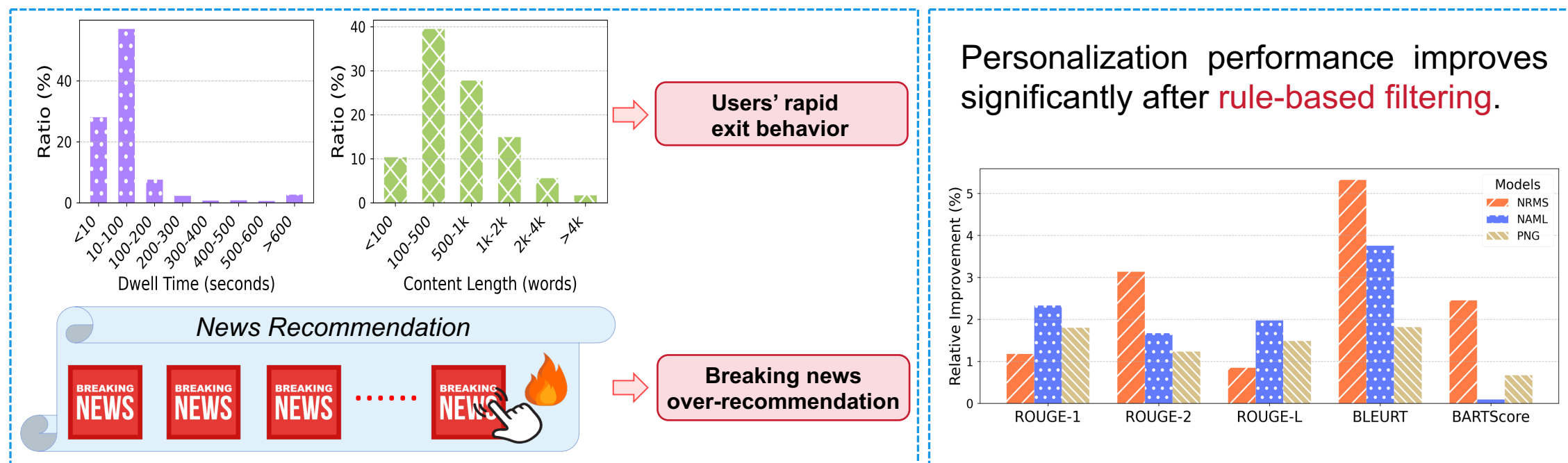
Stephen Curry scored 37 points and dished out 7 assists to lead the Golden State Warriors to a 123-116 victory over the Los Angeles **Lakers**.....
LeBron James put up 33 points and 9 assists for the **Lakers**, and Austin Reaves set a career-high with 9 three-pointers en route to 31 points.



☺ Motivation

Existing methods rely on users' entire click histories but *overlook click noise*.

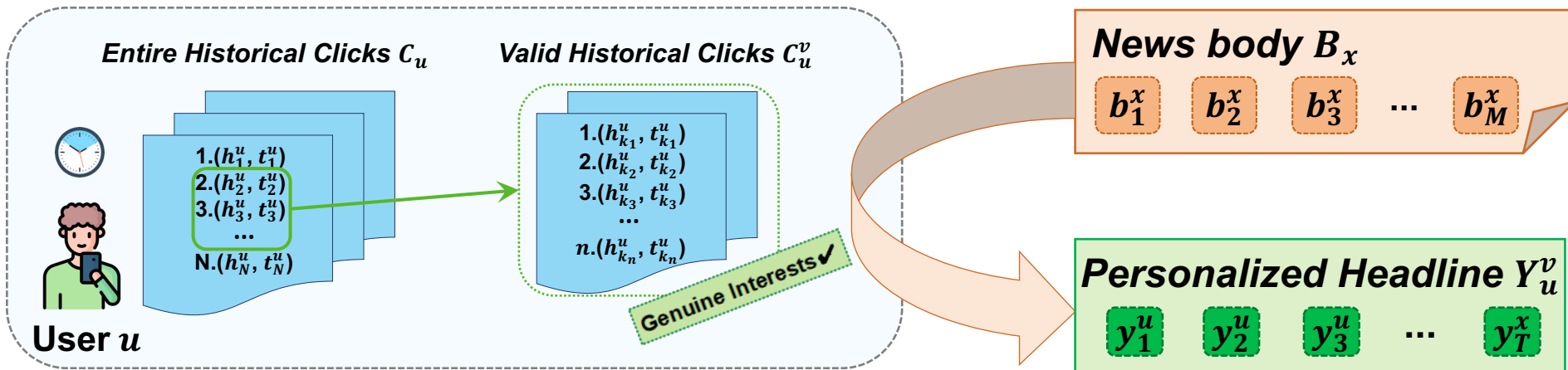
We attribute click noise in personalized user profiles to *the user dimension* and *the news dimension*.



☺ Problem Formulation

Our Goal

- ✓ Construct a *valid* and *noise-irrelevant* historical click sequence.
- ✓ Given a candidate news article, generat a *personalized headline* for the user based on this sequence.



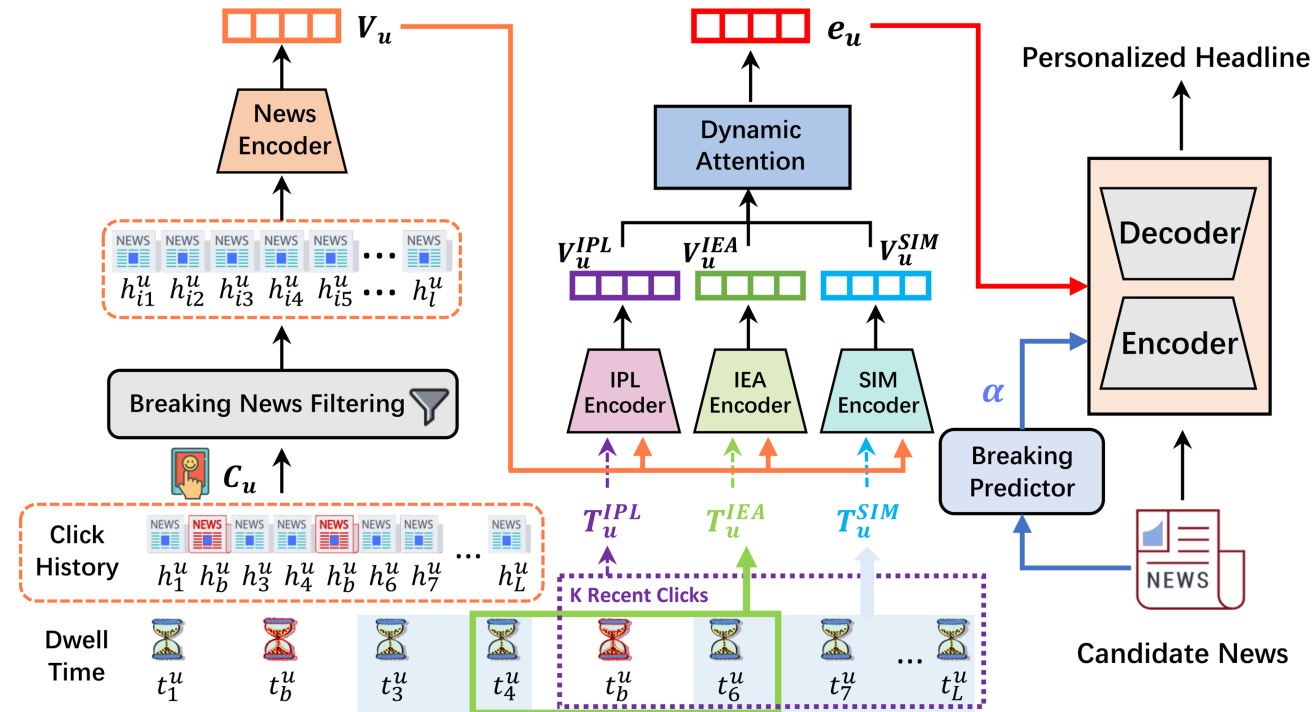
N is the total length of historical clicks.
 n is the total length of valid clicks.

M is the maximum length of the News body.
 T is the maximum length of the headline.

☺ Our Framework

Personalized Headline Generation framework via Denoising Fake Interests from Implicit Feedback

PHG-DIF captures genuine user interests through a **dual-filtering strategy**.

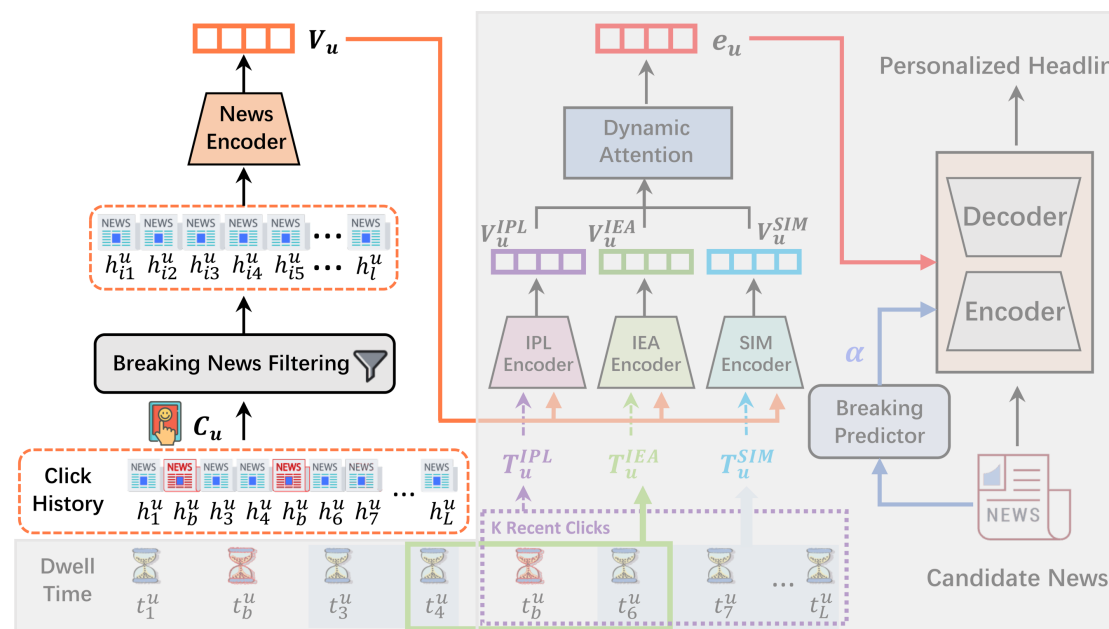


☺ Our Framework

◆ User Modeling with Dual-Filtering

◎ News-Level Filtering

- Construct **breaking news set** by selecting the *top-M* most clicked news headlines from the PENS dataset.
- Mask the **dwell time** of breaking news headlines in the user's historical clicks as zero.



☺ Our Framework

◆ User Modeling with Dual-Filtering

◎ Time-Level Filtering

▣ Instant Preference Learning (IPL)

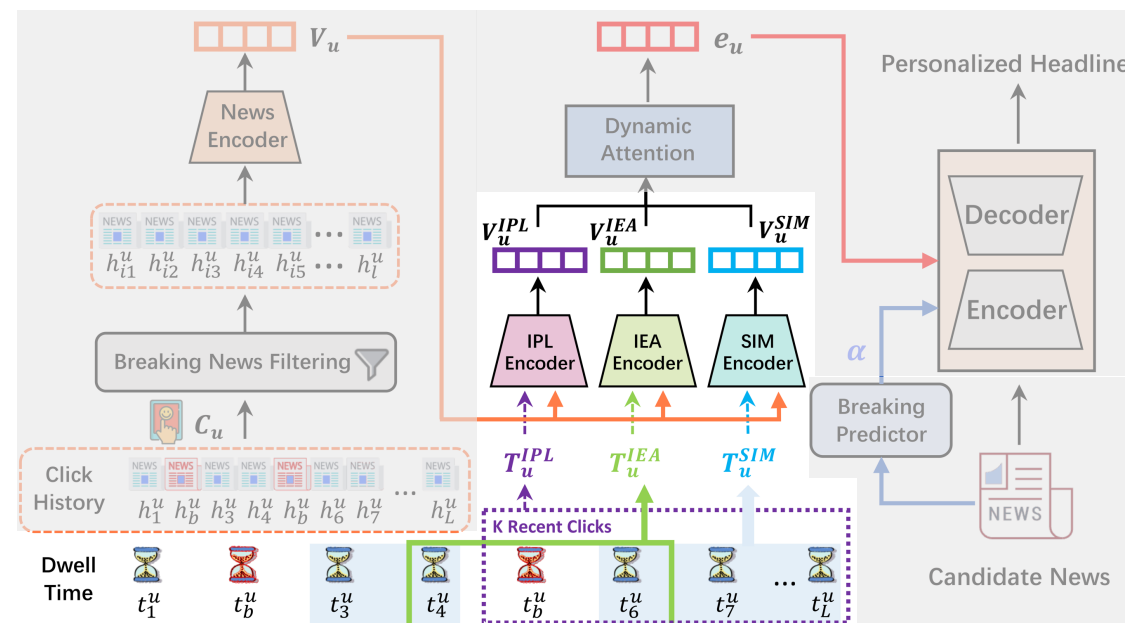
Focusing on the most recent K click histories.

▣ Interest Evolution Analysis (IEA)

Based on click history and dwell times within a certain time window n .

▣ Stable Interest Mining (SIM)

Analyzing click history with consistently high dwell times across their click history.



☺ Our Framework

◆ Multi-Granular Dynamic Aggregation

$$e_u = \text{DynAttn}[v_u^{\text{IPL}}, v_u^{\text{IEA}}, v_u^{\text{SIM}}].$$

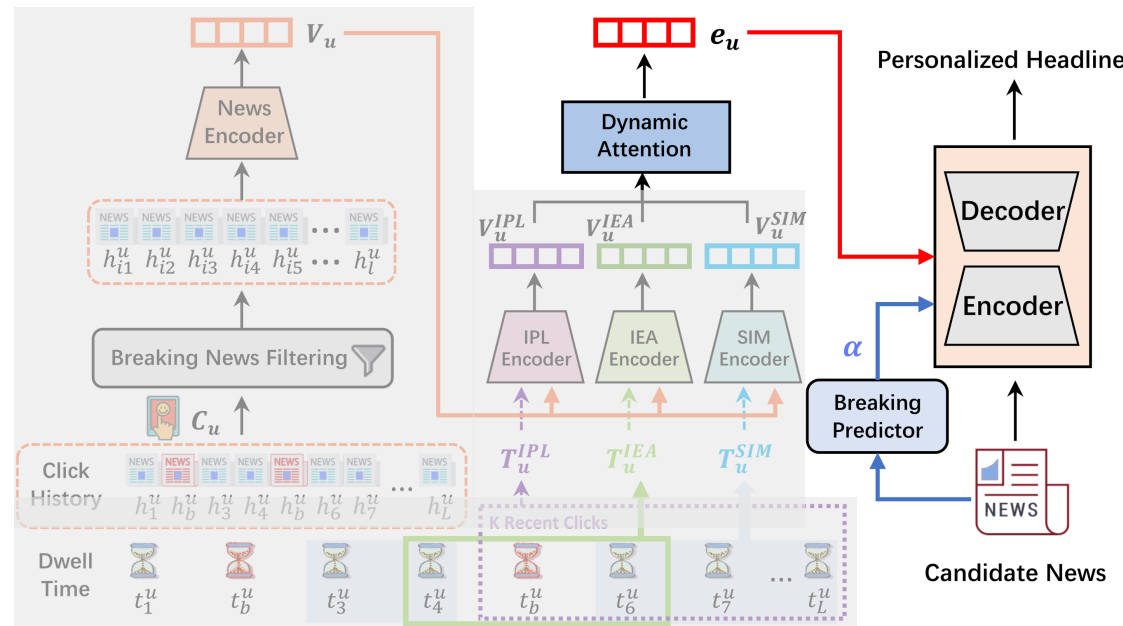
◆ Breaking-News-Aware Generator

$$\alpha = B_\psi(v) \in [0, 1],$$

$$P(w) = \lambda_t P_v(w) + (1 - \lambda_t) \sum_{j: w_j = w} a_{t,j},$$

$$\lambda_t = \sigma(W_\lambda [c_t; s_t; \alpha e_u] + b_\lambda),$$

where a higher α indicates a greater likelihood that the article is breaking news, W_λ and b_λ are learnable parameters and $\sigma(\cdot)$ is the sigmoid function.



☺ Training Schedule

- ◆ Pre-train the encoder on click-through labels.
- ◆ Warm-up the decoder with maximum likelihood estimation (MLE).
- ◆ Fit the breaking-news classifier by binary cross-entropy.
- ◆ Perform a policy-gradient fine-tuning step.

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{Y \sim G_{\theta}}[R(Y)],$$

where $R(Y)$ aggregates several headline-quality indicators.

☺ DT-PENS Dataset

We introduce **DT-PENS**, an extended benchmark with 9823 personalized headlines for 1, 000 MSN users annotated with *historical dwell time*, enabling more robust evaluation of personalized headline generation.

Table 1: Dataset statistics for the original PENS and DT-PENS_{Test} splits. Tokenization was done with NLTK, and truncated at 1,000 words. “wd.” in the table means word.

	Dwell Time	#exposure news	#user
Train	✓	111,762	445,765
PENS_{Test}	✗	60,000	103
DT-PENS_{Test}	✓	23,024	1,000
	avg. click/user	avg. wd./title	avg. wd./body
Train	74.8	10.5	549.0
PENS_{Test}	107.6	10.8	548.2
DT-PENS_{Test}	273.1	9.8	606.6

☺ Overall Performance On DT-PENS

Our proposed PHG-DIF framework *outperforms all baseline methods* across evaluation metrics, demonstrating that *denoising users' historical clickstreams markedly improve personalized headline generation*.

Table 2: Our main experimental results. “-w/o” indicates component ablation (relative % change in parentheses).

Methods	ROUGE-1	ROUGE-2	ROUGE-L	BLEURT	BARTScore
• <i>Non-personalized methods</i>					
BART	18.05	6.70	17.19	26.36	58.03
T5-small	18.90	6.84	17.46	29.60	59.71
• <i>Personalized methods</i>					
PENS-EBNR	23.15	7.19	21.33	44.79	62.28
PENS-NRMS	22.89	7.01	21.15	43.40	60.27
PENS-NAML	23.11	7.17	21.21	44.18	62.20
PNG	23.24	7.28	21.47	45.63	62.29
GTP	24.01	7.58	22.22	48.09	63.80
PHG-DIF_{ours}	24.33*	7.99*	22.47*	48.50*	65.64*
-w/o IPL	24.08(-1.03%)	7.62(-4.63%)	22.23(-1.07%)	48.19(-0.64%)	63.86(-2.71%)
-w/o IEA	24.16(-0.70%)	7.64(-4.38%)	22.29(-0.80%)	48.21(-0.60%)	63.95(-2.58%)
-w/o SIM	23.65(-2.80%)	7.32(-8.39%)	21.76(-3.16%)	46.43(-4.27%)	62.78(-4.36%)

The symbol * denotes the significance level with $p \leq 0.05$. **Bold** font indicates the best-performing method.

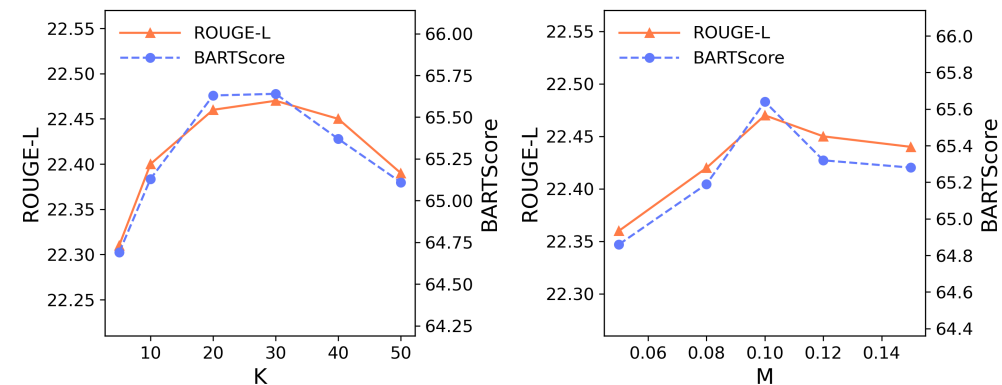
☺ Ablation Study

- ◆ All three encoders are *essential*, SIM has the greatest impact.
- ◆ Removing Breaking News Filtering (BF) or Breaking Predictor (BP) *drops markedly* on every metric.

Table 3: Ablation study results for breaking news handling.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEURT	BARTScore
PHG-DIF (full)	24.33	7.99	22.47	48.50	65.64
-w/o BF	24.12	7.63	22.25	48.20	63.91
-w/o BP	24.29	7.82	22.40	48.41	65.16

- ◆ Overall performance peaks when the IPL window size $K=30$.
- ◆ When $M=0.10$, corresponding to the top 0.10% CTR threshold of all news articles, the overall performance is optimal.



☺ Case Study

PHG-DIF effectively filters out click noise, accurately *capturing the user's true interests* and generating headlines that better *align with user preferences*.

Table 4: A case on personalized headline generation affected by click noise. **Red text** highlights content related to click noise, and **blue text** represents the user’s true interests.

Click History		Dwell Time
British Ambassador to the U.S., Kim Darroch, resigns after Trump criticism		366s
There’s a democratic civil war brewing over decriminalizing migration		115s
A wooden sculpture of Melania Trump was unveiled on the banks of the Sava River		3s
The true cost of high deductible health care plans		7s
Report: Durant, Irving planned to team up before 2018-19 season began		249s
What were the Warriors thinking on Stephen Curry’s final shot?		5s
The Knicks failed to sign Kevin Durant and Kyrie Irving		486s
• Case 1		
Original Headline: Here’s when social security benefits could be cut		
PNG:	Possible impact of Melania Trump’s social security reductions	✗
GTP:	The benefits of Trump’s proposed social security reductions	✓
Ours:	Exploring implications of Trump’s proposed social security cuts	✓
• Case 2		
Original Headline: Calling BS on Stephen A. Smith’s explosive claims on First Take		
PNG:	Stephen Curry calls Stephen A.Smith’s claims about the Warriors	✗
GTP:	Kevin Durant addresses Stephen A. Smith’s claims on his Warriors Era	✗
Ours:	Kevin Durant denies Stephen A. Smith’s claims about his departure	✓

Thanks for listening ~

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The source code will be ready soon at: <https://github.com/liukejin-up/PHG-DIF>.



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GitHub

